Pillar1A2-A: 🤑 Money Supply (M0-M4)

Note- for efficiency in revision, I've split this chapter into two parts: 1A2-A = Money supply M0,M4 etc and 1A2-B = Monetary Policy Tools -CRR, SLR, Repo, OMO, PSL etc.

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11 MONEY'S DEMAND, SUPPLY & CREATION

11.11 DEMAND OF MONEY: LIQUIDITY PREFERENCE THEORY (तरलता अधिमान)

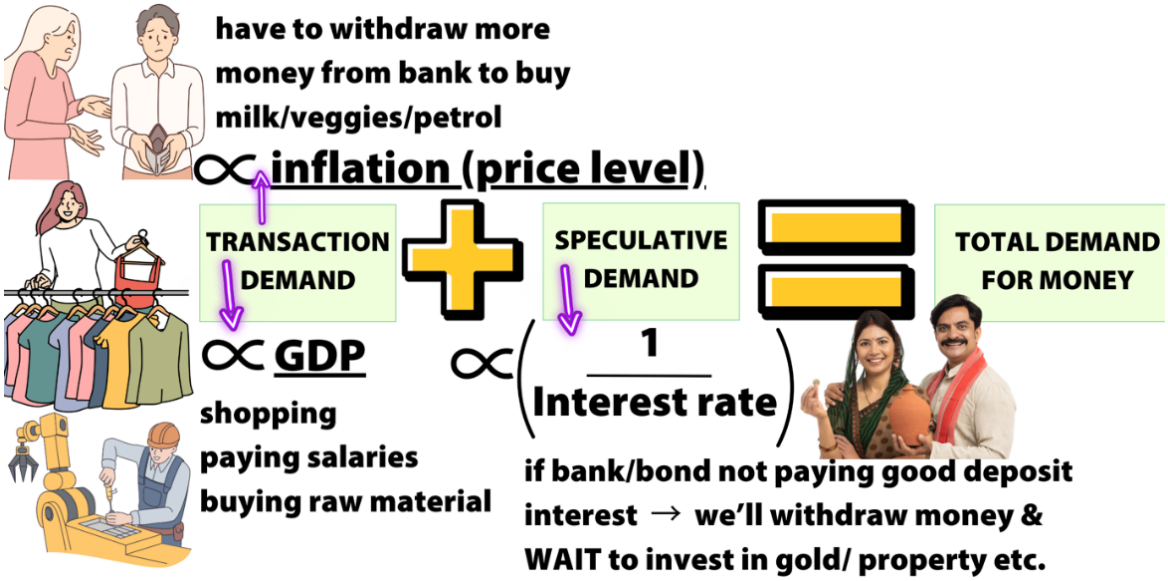
British Economist John Maynard Keynes (Book: *The General Theory of Employment, Interest and Money, 1936*) noted - people prefer to keep a part of assets in liquid form (cash money) with 3 motives: (लोग अपना कुछ पैसा नकदी में हाथ में रखेंगे. सारे का सारा पैसा बैंको में जमा नहीं कर देंगे. क्योंकि तीन कारण होते हैं)

1. **Transaction motive (संव्यवहार उद्देश्य)**: For using money as a medium of exchange e.g. for buying daily milk, vegetables. (रोजाना की खरीद के लिए कुछ पैसा हाथ में रखा जाता है. सारा पैसा बैंक में नहीं जमा करते)
2. **Precautionary motive (एहतियाती)**: To protect against sudden / unforeseen expenditure e.g. medical emergency or impulsive purchase during a holiday trip. (घर में कोई बीमार हो गया, प्रवास में जाना)
3. **Speculative motive (सट्टा-उद्देश्य)**: Investors hold cash to make the best use of any investment opportunity that arises later e.g. waiting for gold / land prices to fall, then "I will use my cash to buy it". Also known as "**Asset Demand of Money**". (जो संपत्ति मुझे खरीदनी है उसका दाम थोड़ा गिर जाए उसकी राह देख रहा हूँ)

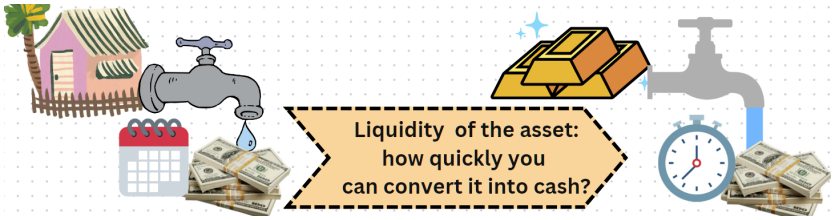
The amount of money held in cash form **vary inversely with (deposit) interest rates**. If higher interest available in Bank Deposits etc. → people would invest money in banks, instead of keeping money in liquid form (cash). (बैंक वाला अगर बचत खाते पर अच्छा ब्याज देगा तो लोग हाथ में नकदी पैसा कम रखेंगे बैंक में निवेश ज्यादा करेंगे)

11.11.1 TOTAL demand of money: factors affecting

- Total demand for money in an economy = transaction demand + speculative demand.
- FAQ: why have you **not** mentioned ***precautionary*** demand in above formula? Ans. Because NCERT has **not** mentioned ***precautionary*** demand in the given formula.



11.11.2 Liquidity of assets

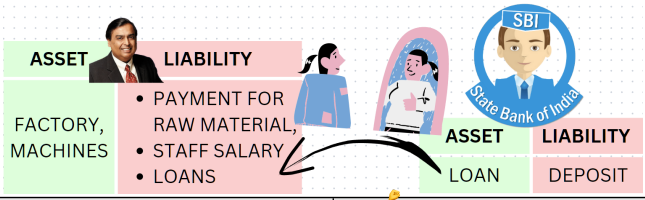


- Liquidity= ease of converting an asset into cash. Cash is the most liquid asset.
- Highly liquid assets = Gold, Demand deposits, G-Sec/T-Bill, shares/bonds of reputed companies. (अत्यधिक तरल संपत्ति: तुरंत खरीदार मिल जाते हैं. आप संपत्ति को नगदी धन में आसानी से परिवर्तित कर सकते हैं)
- Relatively illiquid assets= Home/Real estate, Paintings/Sculptures etc. Because difficult to find buyers at right price instantly. खरीदार आसानी से नहीं मिलता इसलिए इन संपत्तियों की तरलता थोड़ी कम है
- Liquidity injection / infusion refers to phenomenon when RBI buys Bank/NBFCs' G-Sec/T-bill/financial assets to give them cash. (HOW? Ans. Watch Video) (तरलता अन्तःक्षेपण)

11.11.3 Liquidity Trap (तरलता जाल) – refer to 1A2-B Handout.

- Refer to the 1A2-B Handout.

11.12 Deposit Types



Time Liabilities of a Bank (FDRD) समय / मीयादी देयताएँ	Demand Liabilities of a Bank (CASA) मांग देयताएं
- Fixed deposits, Cumulative/ recurring deposits, Staff security deposit etc. - Bank legally not required to pay customer before maturity. But bank may pay after deducting penalty/ interest. (वक्त से पहले पैसा निकाला तो दंड लगेगा)	- Current Account , Savings Account - Demand Draft - Overdue balance in Fixed Deposits - Unclaimed deposits. (What does it mean? Ans. Explained in Video Lecture.) - What is Nostro/Vostro? Refer Pillar3B.
Public parks more money here, because better returns / higher interest rates.	LESS (ऐसे बचत खातों पर ब्याज कम मिलता है इसलिए जनता कम मात्रा में पैसा रखती है यहाँ पर)
LESS liquid.	More liquid bcoz easily convertible into cash

____ is not included in the assets of a commercial bank in India? (Prelims-2019)

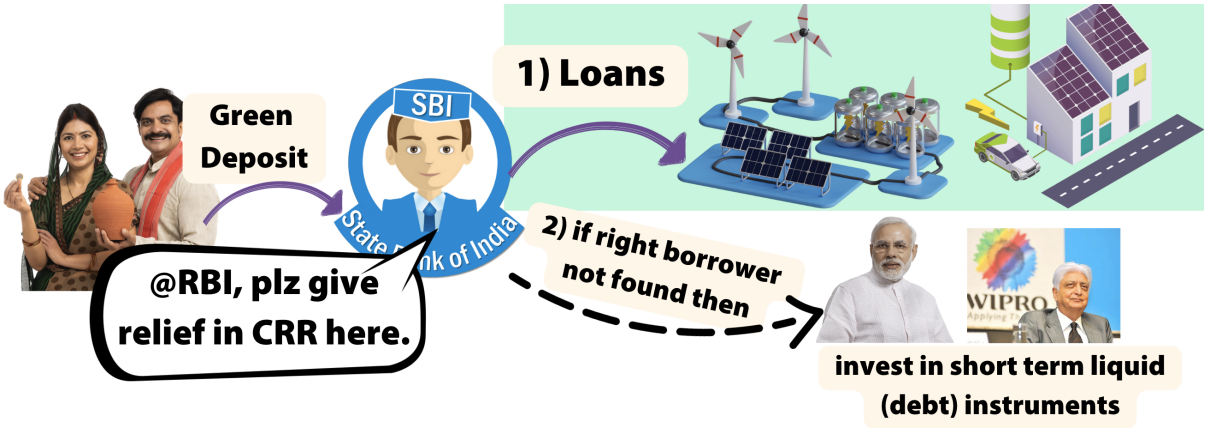
(a) Advances (b) Deposits (c) Investments (d) Money at call and short notice

11.12.1 DEPOSITs → Savings Account → Average Monthly Balance ELSE penalty

- The average monthly balance (AMB) is the minimum savings account balance required by banks. Bankers impose penalties for falling below the AMB.
- 2025: ICICI Bank hiked to the minimum balance rule ➡ public outrage. Meanwhile SBI,

PNB, Canara etc. removed such penalties.

11.12.2 🏦🍀 DEPOSITs → Time Deposit → Green Deposit (2023)



- 2023: RBI released guidelines. It is a Type of fixed deposit. This deposit money is loaned to green / environment related.
- Who can collect such deposits? banks, deposit-taking NBFCs (e.g. Bajaj Finserv), Housing Finance companies (e.g. Canfin Homes)

✅ Banker given give loans to	✋🚨 loans not allowed in
renewable energy, waste management, clean transportation, energy efficiency, and afforestation	fossil fuels, nuclear power^^, tobacco, etc business.

^^🚩👶 **FAQ why RBI forbidden it in nuclear power? Even nuclear power is clean energy!?** Ans. RBI governor didn't bother to justify in press release so, Plz email your suggestions to governor@rbi.org.in if you have a lot of free time.

- SBI launched Green FD with 3 tenures — (A) 1,111 days (B) 1,777 days (C) 2,222 days.
- If such bank/NBFC unable to find green-borrower in immediately (for giving long term loans) → then bank/NBFC can invest such green deposit money into liquid instruments with max. maturity upto 1 year **e.g. T-Bill, Commercial Papers etc.** (Ref Pillar1C for more.)
- If such bank/NBFC accepted green deposit but can't find green-borrowers, then RBI will not penalise bank/NBFC for it.
- These deposits are covered by Deposit Insurance and Credit Guarantee Corporation (DICGC). So even if the bank fails/shuts down → depositor will get upto ₹5 lakh refunded by DICGC. (Ref Pillar1B2 for more.)
- Green deposits must be issued/accepted in rupee currency only. (i.e. not in dollar/Euro etc.)
- It is not compulsory every Bank/NBFC to raise/accept green deposits. Unki marzi hai. (just like it is not compulsory for every automobile company to launch an E-Vehicle.)

🚩👶 *FAQ: Do I've to remember all this? You added almost one page summary on Green Deposits, and so now your handout will become even longer!? Do I have to remember this much?* *Ans. Yes. Bcoz (A) This Green Deposit topic also relevant for the* ***environment****. (B) UPSC examiners new bad habit of asking microscopic facts of schemes.(C) I'm not pointing a gun over ur head to read it.*

11.12.3 Time Deposit → Fixed Deposit → Bulk Deposits (₹3cr/>)

Definition given by RBI	before	After-2024
Bulk Deposit means a large sized fixed deposit of _ _ _	₹2 cr or higher	₹3 cr or higher

- Bankers may offer different interest rate on bulk deposit compared to normal fixed deposits.
- Whether interest will be higher or lower than the normal deposit depends on the banker's business situation. We will not Phd on this. (इसपे बेंकर सामान्य एफ़डी से कम/ज़्यादा ब्याज दे सकता है।)

11.12.4 Demand deposits → Savings account → Salary account

e.g. If Infosys has 5 lakh employees, Infosys will open their savings accounts in a bank & deposit the monthly salary inside it. Such accounts are called salary account. Further banking GK NOTIMP.

11.12.5 Deposits→ Inoperative & unclaimed accounts

Inoperative Account	unclaimed deposits
If depositor not doing any transaction from his side for more than 2 years	(1) if savings/current account are not operated for 10 years (2) if fixed/term deposits not claimed within 10 years from date of maturity.
Over ₹ 1 lakh cr stuck here (2023 data)	Over ₹ 42,000 cr stuck here (2023-data)
Bank can't charge penalty on such customer if it is a savings account, then banker must continue paying interest on it.	same as left cell.

11.12.6 : UDGAM Portal: Unclaimed Bank Deposits, Awareness Fund (2023)

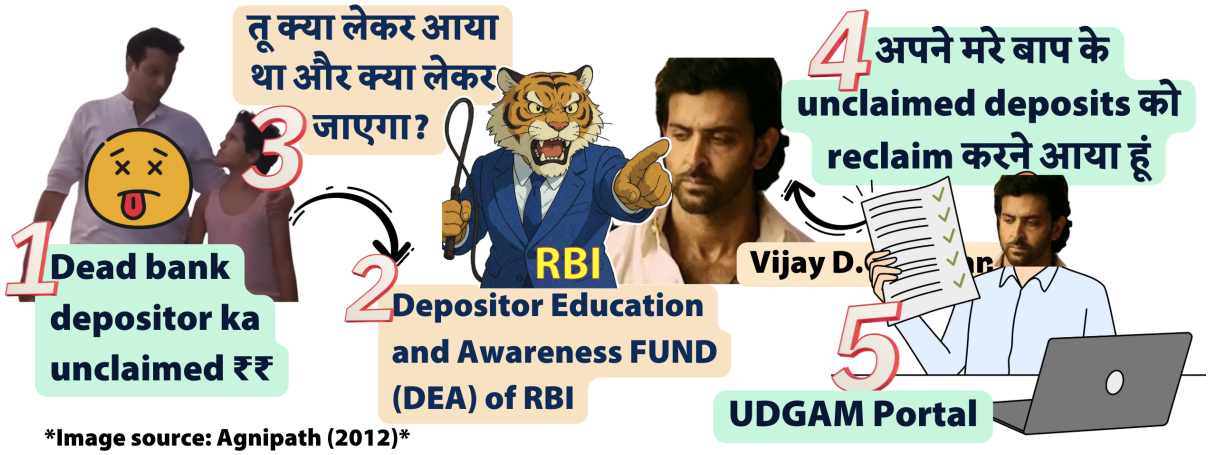
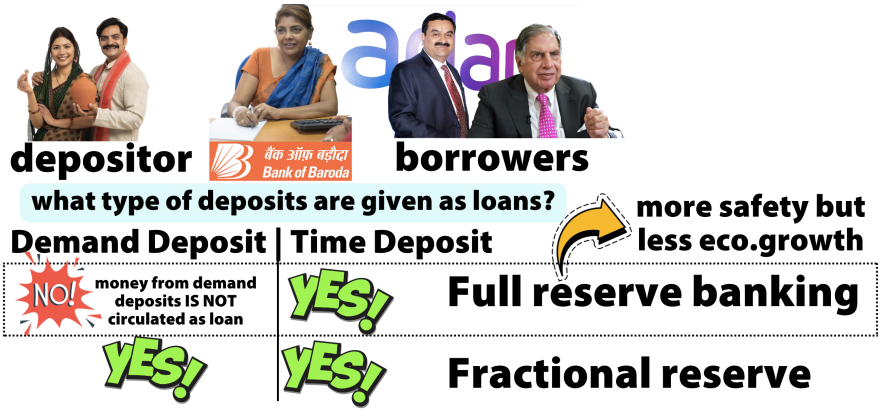


Image source: Agnipath (2012)

- If deposits remain unclaimed for 10 years in a bank → banker need to transfer it to "Depositor Education and Awareness FUND" (DEA) OF RBI. 2023: RBI setup UDGAM portal for it.
- Beneficiary can easily search unclaimed deposits across multiple banks at one place. (e.g. by searching the Aadhaar card / PAN Card number of his/her dead parent etc.)

11.12.7 Full Reserve Banking vs. Fractional Reserve Banking

> इन शब्दों का हिंदी अनुवाद नहीं करेंगे क्योंकि इस प्रकार की थियरी पूछने का मुख्य परीक्षा में कोई रिवाज नहीं है.



	Full Reserve Banking	Fractional Reserve banking
Can banker give money from DEMAND Deposits as LOANS? (e.g. current account, saving account) →	NO. And consequently, Banker may not pay saving interest to demand depositors.	YES, can give.
Can banker give money from TIME Deposits as LOANS? (e.g. Fixed Deposits) →	YES, can give.	YES, can give
Bankers' vulnerability to sudden/panic withdrawals by clients (known as 'Bank Run')	LESS. Because demand deposits not given as loans.	MORE. Because both time and demand deposits are given as loans. So if Vijay Mallya fled the country → it will create more problems!
Banker's ability to help in Economic growth?	Less. Because banker giving less qty of loans.	More

- **Note:** India uses Fractional Reserve Banking system.

11.13 Measures of Money Supply (मुद्रा आपूर्ति के परिमाण)

- (Objective) To design a good nutrition scheme for India, the government will have to first find the weight of the citizens. Similarly, to make a good monetary policy, RBI will have to find the quantity of money in the system. (कुपोषण के ख़िलाफ़ अच्छी योजना बनाने से पहले सरकार ने लोगों का वज़न नापना होगा. उसी तरह देश के लिए अच्छी मौद्रिक नीति बनाने के लिए रिज़र्व बैंक ने सबसे पहले तो देश में पैसों का जत्था नापना होगा)
- (**Definition**) Money supply means the total amount of money in an economy at any given time. (दिए गए समय पर अर्थतंत्र में कुल मिलाकर कितना पैसा है उसकी मात्रा या जत्थे को कहते हैं "मुद्रा आपूर्ति")
- (**Significance**) Money supply plays a crucial role in 1) price level (=inflation) and 2) interest rates on deposits & loans. (मुद्रा की आपूर्ति → महंगाई दर और ब्याज दरों पर असर करती है)
- (**Types**) RBI measures the money supply through indicators: M0, M1, M2, M3, M4.
- See Next Page for diagram and tables

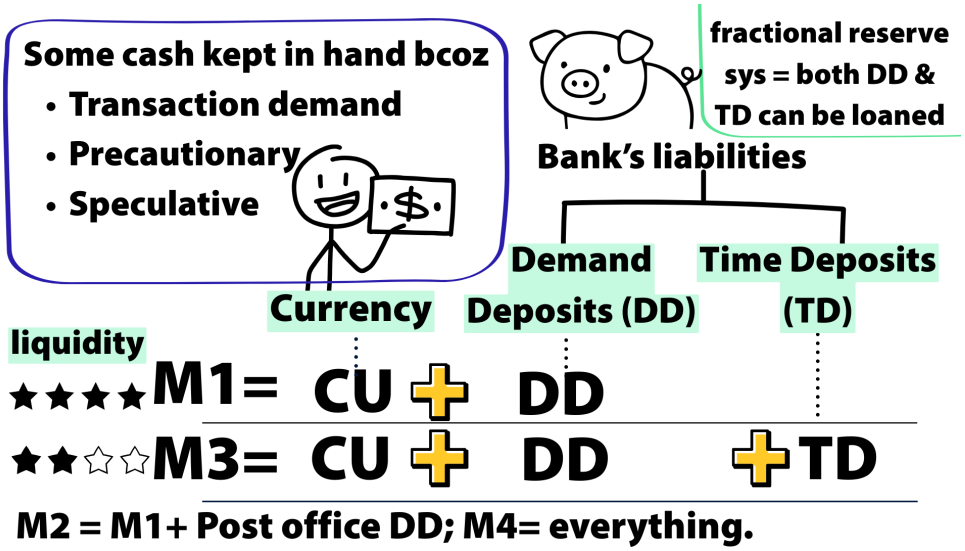


*Table 1: *CU: Coins & Currency with Public*

Measure संकुचित, व्यापक		*CU	Commercial Banks		Post Office Savings Bank		Liquidity तरलता	Qty मात्रा
			Demand Deposits (CASA)	Time Deposits (FDRD)	Demand Deposits (SA)	Time Deposits		
Narrow Money	M1	✓	✓	✗	✗	✗	♠♠♠♠	♠
	M2	✓	✓	✗	✓	✗	♠♠♠	♠♠
Broad Money	M3	✓	✓	✓	✗	✗	♠♠	♠♠♠
	M4	✓	✓	✓	✓	✓	♠	♠♠♠♠

Apply logic to derive formula e.g. M3 = M1 + Time deposits with banks.

- M3 is the most commonly used measure of money supply, also known as **Aggregate Monetary Resources / Aggregate Money Supply**" (समस्त मौद्रिक संसाधन) because out of all the money supply indicators (M0-M4) this is the indicator RBI will focus the most for its analysis while designing of monetary policy. (अपनी मौद्रिक नीती विश्लेषण के लिए रिज़र्व बैंक इस आंकड़े पर खास ध्यान रखती है)
- In above formulas for money supply, we are only counting the "**NET Demand / NET Time deposits**" i.e. only public's deposits in bank. We are not counting inter-bank deposits i.e. one commercial bank's deposit in other commercial banks. (एक बैंक से दूसरे बैंक में पैसा डिपॉजिट किया हो तो उसे इधर नहीं गिनते.)
- **M1 and M2 are called Narrow money.** Because of their smaller size (since they only cover demand deposits) M3 and M4 are called broad money, because of their relatively larger size than M1 and M2. since these terms are given in the NCERT so we have to prepare & accept. It is not some great thing worthy of mains exam that we have to spend more time than that 1 word association.
- 🚩👶🏻**FAQ: What is Hard Money vs Soft Money?** wait till we reach Pillar3B currency exchange.
- 🚩👶**FAQ**: *As per NCERT class12: "M4 = M3 + Total deposits with Post Office savings organisations (excluding National Savings Certificates)". So why you revision table doesn't talk*

about National Savings Certificates. ✅ Ans. This much pedantry I find it less imp for exam so don't want to overcomplicate my revision friendly table.

- 🚩👶 **FAQ1**: Why M3 is Aggregate Money Supply? ✅ Ans. Watch video.
- 🚩👶 **FAQ2**: Why is the liquidity of M4 less than M3? ✅ Ans. Watch video.
- 🚩👶 **FAQ3**: What is Narrow Money and Broad Money? ✅ Ans. Watch video.

🔠❓ **MCQ. If you withdraw Rs. 1 lakh in cash from your Demand Deposit Account at your bank, the immediate effect on aggregate money supply in the economy will be _ _ . (Pre-2020)**

[a] to reduce it by ₹ 1,00,000 [b] to increase it by ₹ 1,00,000
[c] to increase it by more than ₹ 1,00,000 [d] to leave it unchanged

🚩👶 FAQ: I've 500 jaat-ke doubts in above MCQ ✅ Ans. Watch lectures.

🔠❓ **MCQ. Which of the following are included in Narrow Money? (CDS-2023-II)**

1. Currency with the public 2. Demand deposits
3. 'Other' deposits with Reserve Bank of India 4. Banker's deposits with Reserve Bank of India

Codes: (a) 1, 2 and 4 (b) 1 and 2 only (c) 1, 2 and 3 (d) 3 and 4 only

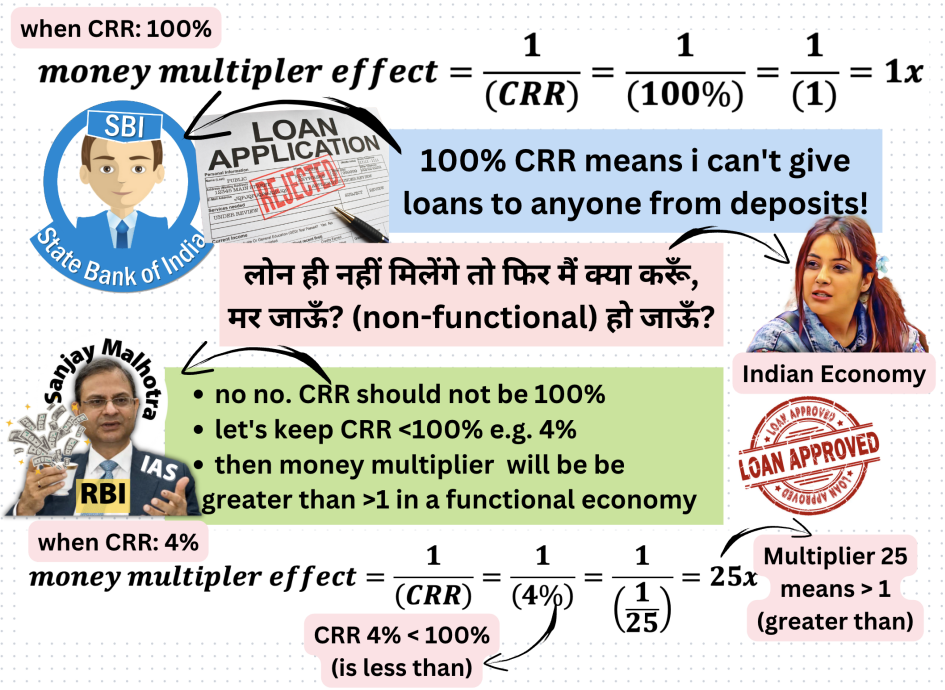
11.13.1 ⏰ Call Money | Notice Money | Term Money

These terms are used when bank/Non-Banking Financial Companies (NBFC) lend/borrow money among themselves for short term (i.e. less than 1 year)

word	Duration of LOAN
Call Money	borrowed for one day (=This duration also called "overnight".)
Notice Money	2 days to 14 days
Term Money	Exceeding 14 days but less than 1 year

✋DONOT mix above words with broad money / narrow money / M1/M2/M3/M4 etc.

11.14 💸⚡⛵ Money Multiplier (मुद्रा गुणक)



- Money Multiplier (MM) refers to the amount of M3 a bank generates with each rupee of M0 available to them.
- MM depicts the relationship between the monetary base (M0) and the economy's money supply (M3), indicating how fast the money supply will grow from the bank's lending activity.
- Higher the value of the MM, the higher the liquidity in the market and vice versa.
- RBI's Cash Reserve Ratio (CRR) controls the ***Fractional Reserve Banking & Credit Creation*** by the commercial banks, in the following manner:

Table 2: this process is called 'Fractional Reserve Banking & Credit Creation'

High Powered money (M0) = 100	**Deposits**	**Loaning**	**10% Reserve ('R')**
Bank#1	100	90	10
Bank#2	90	81	9
Bank#3	81	72.9	8.1
Bank#..	..	..	..
Total Money (here M3)	Total sum = ₹1000		

$$\textbf{\textit{Money Multiplier}} = \frac{Stock\ of\ Total\ Money\ (M3)}{Stock\ of\ High\ Powered\ money (M0)} = \frac{1000}{100} = 10$$

- In above example, 10% CRR (CRR) generated 10x times of M0 money. [बैंक कर्ज देने से पहले यदि 10% रकम आरक्षित रखती है तो 10 गुना ज्यादा पैसों का निर्माण]
- 4% reserve ratio will generate [1/(1/25)]= 25x times (in theory). See formula below:

$$CRR\,4\% = \frac{4}{100} = \frac{1}{25}$$

$$\textbf{\textit{money multipler effect}} = \frac{\mathbf{1}}{(\boldsymbol{CRR})} = \frac{\mathbf{1}}{(\mathbf{4\%})} = \frac{\mathbf{1}}{(\frac{\mathbf{1}}{\mathbf{25}})} = \mathbf{25\ \textit{times}}$$

- If CRR is 4% then MM should be 25x theoretically. ✋ however in reality the multiplier may be lower than 25x, due to poor banking penetration. (गांव गांव तक बैंक की सुविधा नहीं है इसलिए वास्तविक जीवन में इतना मुद्रा गुणक नहीं होगा)
- In a functional economy, money multiplier (MM) is always greater than 1 & CRR will always be less than 100%. **WHY? Ans. Watch video.**
- MM directly improves with ↓reduction in CRR. (सीआरआर घटेगा तो मुद्रा गुणक बढ़ेगा)
- MM indirectly improves as economy develops, consumption / loan demand increases, banking penetration, digital economy, less-cash economy etc.
- MM improves with the ↑in banking penetration, financial inclusion (वित्तीय समावेशन)
- Boom period, Whenever loan demand ↑ (तेजी का माहौल → लोन ज्यादा लिए जा रहे हो)
- When RBI adopts Cheap / Easy / Dovish / Expansionary monetary policy to combat deflation. (जब सस्तापे/ अपस्फीति से लड़ने के लिए रिज़र्व बैंक लोन के ब्याज दर सस्ते कर दे)
- In 1960s = less than 2x, 90s = more than 3x, At present = more than 5x.

🚩👶 FAQ: What will be the impact of SLR changes on Money Multiplier? Ans. Can't be written mathematically. Reason given in Video.

11.14.1 Money Multiplier (MM)- similarities with boat-speed to prevent confusion

Speed = Distance / Time	**Money Multiplier = M3 / M0**
wind can increase/decrease boat speed	CRR can increase/decrease Money Multiplier (MM)
Speed of boat also affected by 1. Weight of Cargo 2. size of the sail-cloth	**MM also affected by** 1. Income & saving of public 2. Banking penetration / financial inclusion 3. Deposit-Loan activity etc

11.14.2 CRR & Cash are 'leakages' to money multiplier says ES25

The money multiplier (MM) = how much money a banking system can generate from each unit of central bank money (M0). MM mainly depends on two factors:

Factor	Implication
1. Cash Holdings	More cash held by individuals = money leaked from bank deposits= MM ⇓⇓
2. CRR	Higher CRR = banks hold more reserves with the RBI means less loanable funds with the banks → MM ⇓⇓. Thus, CRR also acts as a leakage.

11.14.3 Money Multiplier From 1981 to 2020

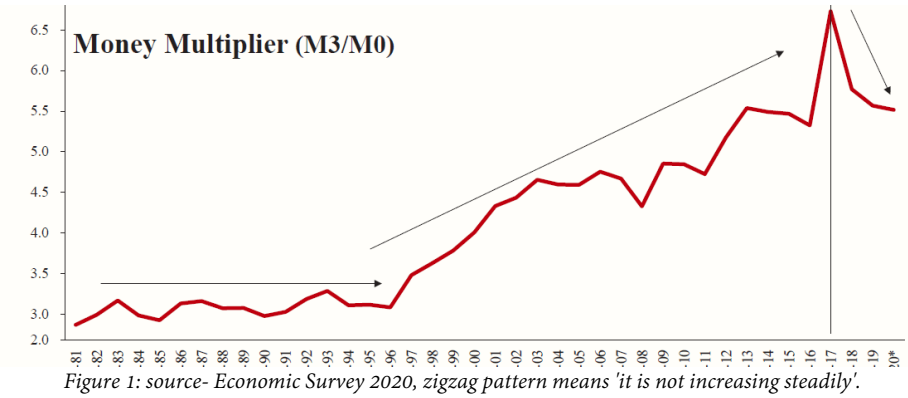
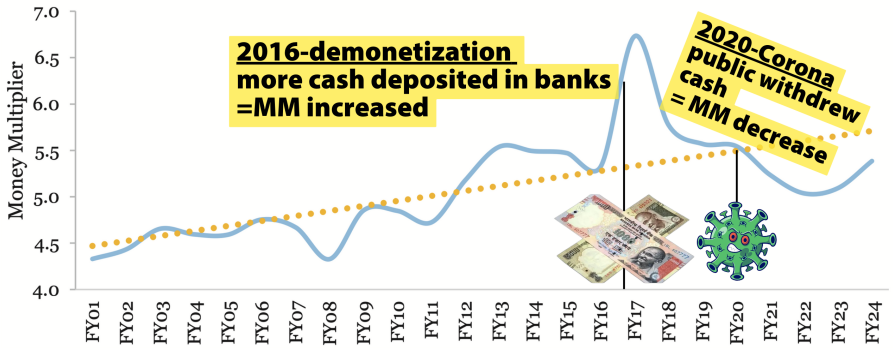


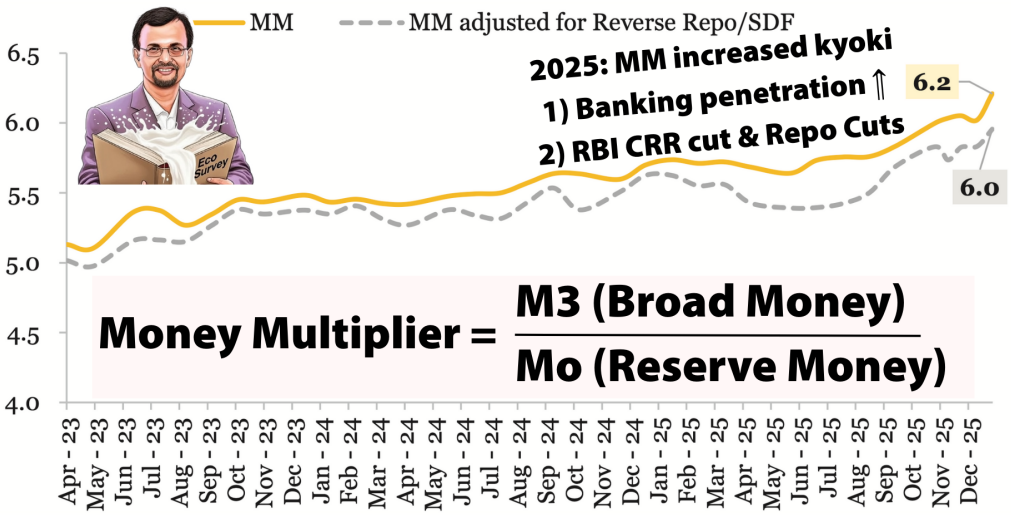
Figure 1: source- Economic Survey 2020, zigzag pattern means 'it is not increasing steadily'.

- between mid-1990's to 2016-17: Money multiplier (measured as a ratio of M3/M0) was mostly increasing↑ due to LPG reforms & increased banking penetration.

11.14.4 💸⚡📘ES25: Money Multiplier From 2001 to 2024



11.14.5 💸⚡📕ES26: M0-M3 Trends & Money Multiplier



🔠❓ ***MCQ. Money Multiplier in an economy increases with _ _ _? (UPSC-Prelims-2019)***

A) Increase in the cash reserve ratio B) Increase in the banking habit of the population

C) Increase in the SLR D) Increase in the population ☝***Same MCQ repeated in Prelims 2021***

MCQ. What would be the impact on the economy if people start holding more currency in hand and less in deposits? (CDS-2023-II) *(a) Money demand will increase (b) Money multiplier will decrease (c) Money multiplier will increase (d) Money demand will decrease*

11.14.6 (Clarification) Formula for Money-supply M0-M4 not matching with CDS Exam

- M0 to M4 formula in NCERT are different than real life formulas in RBI website/Eco Survey.
- In the past, 1 formula MCQ in CDS- its answer-keys not matching with NCERT, because nobody filed a complain to UPSC after the exam was over within seven days. at this link https://upsconline.nic.in/miscellaneous/QPRep/
- if CDS-candidates had filed complain, then UPSC may have cancelled the MCQ.
- Note: I'm not giving that controversial MCQ here, to prevent unnecessary mental stress.

Formulas from Economic Survey 2024 (IGNORE THIS. REMEMBER NCERT formulas)

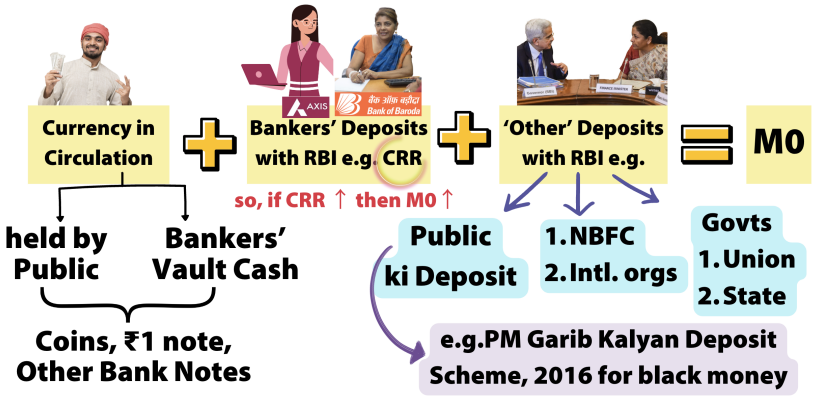
(1) Components of Broad Money (M3) =Currency with the Public + Aggregate Deposits (Demand Deposits with Banks + Time Deposits with banks + 'Other' deposits with Reserve Bank)

(2) Sources of Broad Money (M3) =Net Bank Credit to Government + Bank Credit to Commercial Sector + Net Foreign Exchange Assets of Banking Sector + Government's Currency Liabilities to the Public- minus Banking Sector's Net Non-Monetary Liabilities)

11.14.7 M0 formula: NCERT vs RBI (Also called Reserve Money / high powered money)

NCERT formulas	RBI & Economic Survey formulas
M1 = CU + DD M2 = M1 + Savings deposits with Post Office savings banks M3 = M1 + Net time deposits of commercial banks M4 = M3 + Total deposits with Post Office savings organisations (excluding National Savings Certificates) where, CU is currency (notes plus coins) held by the public and DD is net demand deposits held by commercial banks. The word 'net' implies NCERT Textbook page 38–39 Ref: https://ncert.nic.in/ncerts/l/leec103.pdf **My handout contains NCERT formulas.** **I see no benefit in explaining / memorizing the RBI/Eco Survey's official technical formula-** from exam point of view. You'll be overwhelmed and struggle to memorize them anyways. better skip it, and use brain memory for other topics. If you see some time-pass private mock test-wala making MCQs out of these official RBI/Eco Survey formulas, then you should ignore such MCQs, for your own mental peace.	**RBI official formula (IGNORE THIS. and REMEMBER NCERT)** Reserve Money = Currency in circulation + Bankers' deposits with the RBI + 'Other' deposits with the RBI = Net RBI credit to the Government + RBI credit to the commercial sector + RBI's claims on banks + RBI's net foreign assets + Government's currency liabilities to the public – RBI's net non-monetary liabilities M_1 = Currency with the public + Demand deposits with the banking system + 'Other' deposits with the RBI. M_2 = M_1 + Savings deposits of post office savings banks M_3 = M_1 + Time deposits with the banking system = Net bank credit to the Government + Bank credit to the commercial sector + Net foreign exchange assets of the banking sector + Government's currency liabilities to the public – Net non-monetary liabilities of the banking sector M_4 = M_3 + All deposits with post office savings banks (excluding National Savings Certificates).

11.15 M0: FORMULA WIT COLOURFUL GRAPHICS

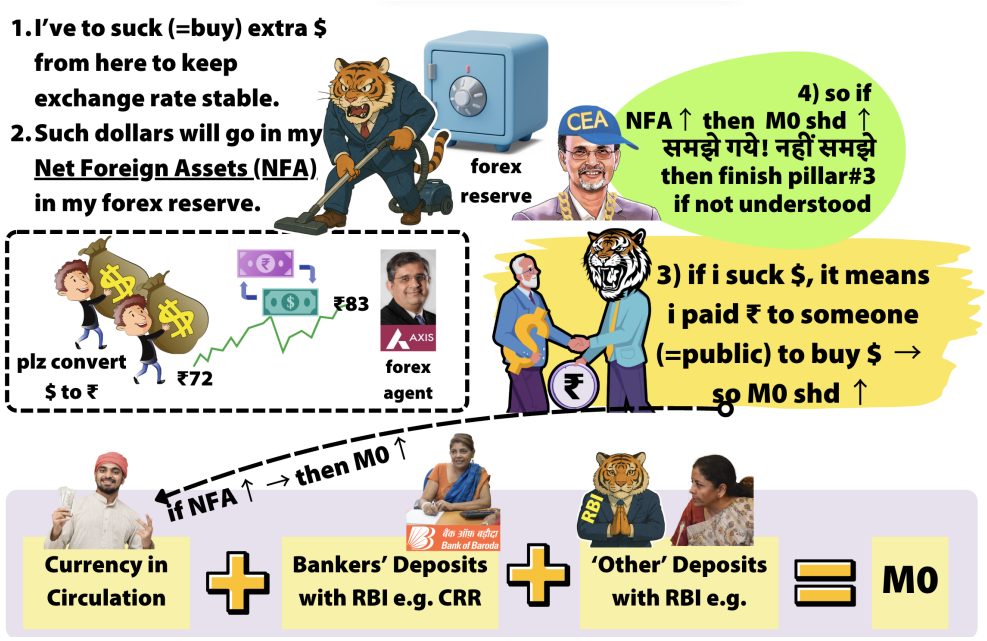


- M0 measures the total amount of paper money and coins in circulation, plus the reserves (deposits) of central bank (=RBI).
- It is called M0 / Reserve Money / Monetary base / Govt Money / High Powered Money.

- Different books give different answer. Basically it boils down to image given above.
- **"Vault cash"** = cash kept by bankers with themselves, for day to day operations.

11.15.1 If CRR ↑ then M0 ↑ says Eco Survey

How? Ans. Refer to above image and read the red fonts.



11.15.2 If net foreign assets (NFA) ↑ then M0 ↑ says Eco Survey

Ans. Explained in previous chart. It requires prior understanding of the currency exchange topic from pillar#3. So revisit the following chart after finishing pillar#3.

11.16 M0: Creation of Money (मुद्रा का निर्माण)

Watch Video lecture before reading further. ELSE u'll have too many doubts.

is issued under RBI Act, by RBI's ISSUE DEPARTMENT, with condition that ISSUE DEPARTMENT's assets must match its liabilities. (आरबीआई केवल उतना ही पैसा छाप सकता है जितनी परिसंपत्तियां उसके इश्यू-विभाग के पास हो)

Assets of Issue Department (परिसंपत्तियाँ)	Liabilities of Issue Department [M0]
1. Rupee coins [RBI 'buys' coins & ₹1 notes from Govt and circulates it as the 'Agent of govt'] 2. Gold coins [Min. ₹ 200 crores] सोने के सिक्के 3. Gold bullion [Min. ₹ 115 crores] सोने की ईंटे 4. Foreign Securities, incl. IMF [Earlier Minimum ₹ 400 crores but Post-1995 no such requirement.] 5. (Indian) Govt. Securities [through which Govt borrows money & returns Principal + Interest at later date] सरकारी प्रतिभूतियां	Total Bank notes in circulation, which consists of: 1. Currency in circulation (i) Held by Public (ii) Held by other Banks in "Vault cash" (i.e. amount kept by bankers for day to day Operations.) 2. Bankers' Deposits with RBI (e.g. CRR) 3. Other deposits with RBI (of Public*, NBFCs, & Govts, Intl.org)

Assets of Issue Department (परिसंपत्तियाँ)	Liabilities of Issue Department [M0]
Total = suppose 100 Cr @"X"date	Total = 100 Cr @"X" date.

- * such as PM Garib Kalyan Yojana (2016) under which Blackmoney holder was required to deposit 25% of his blackmoney in RBI for a lock-in period of 4 years. More in Pillar#2B
- M0 ↑ increases when RBI's asset side increases ↑ . (जब सरकार रिज़र्व बैंक से ज्यादा पैसा कर्जे मे ले तब रिज़र्व बैंक की परिसंपत्तियों में बढ़ोतरी होती है)

11.16.1 FAQ on Rs.1 paper-note

FAQ: Whose responsibility is it to exchange the public's torn / soiled ₹ 1 note ?

Ans. RBI will do it as the agent of the Govt.

FAQ: Whose responsibility is it to exchange broken coins?

Ans. First of all it is difficult to break a coin unless with a metal cutter. And secondly I'd not loose sleep over such JAMES-Bond-Babu-Shona Qs. for UPSC exam point of view. Common-sense suggests that if RBI is exchanging torn ₹1 Note, then RBI would be exchanges coins as well.

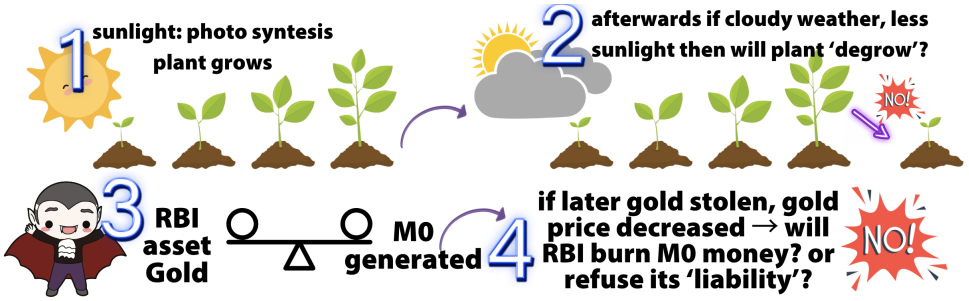
FAQ: All coins and ₹1 paper-note is whose liability ? RBI or Govt

⇒ All coins and ₹1 paper-note are issued by Government. (Ref: Pillar1A1 Handout)
⇒ RBI circulate them as "agent of the government".
⇒ However these items are liability of government. As per RBI Website.
⇒ So, All coins & ₹1 paper-note are liability of Govt.
⇒ However, some books interpret that since 1) RBI is circulating them 2) RBI is responsible for getting torn ₹1 rupee notes exchanged, so it is RBI's liability.
⇒ If in Doubt for MCQ → then tick "Govt." as Answer.

11.16.2 FAQ: How can same thing become Asset and liability ? (WATCH VIDEO)

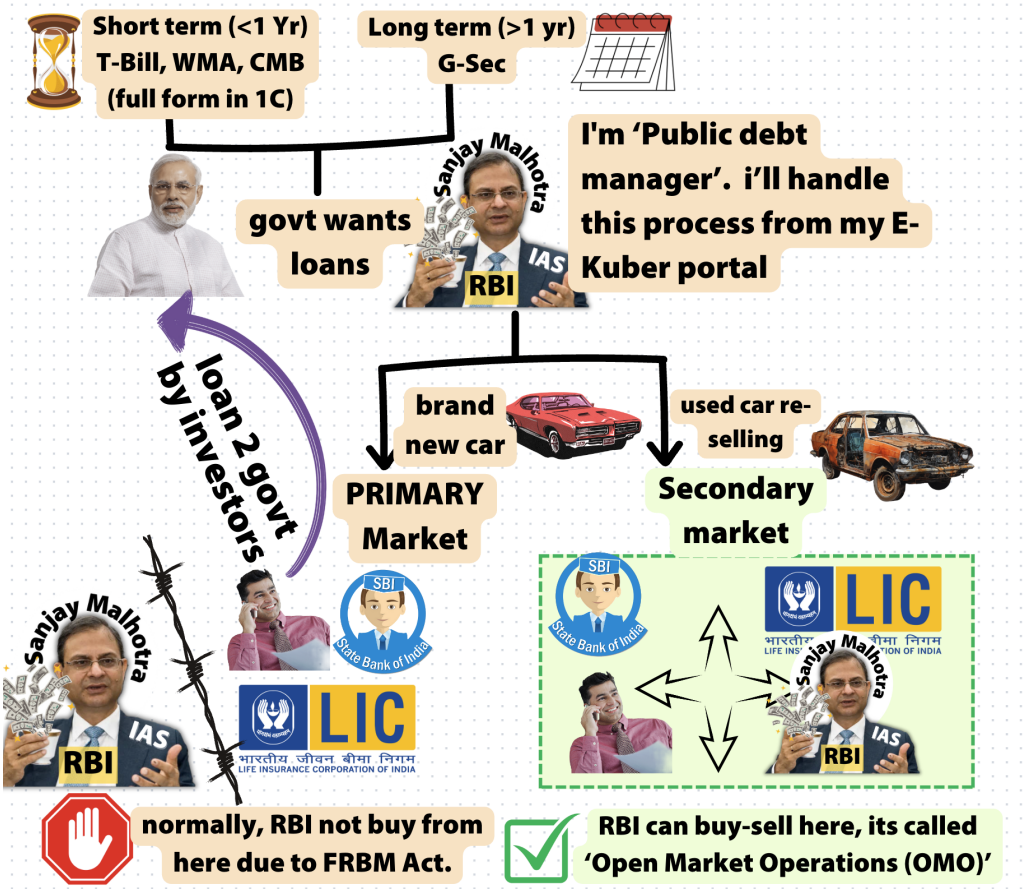
11.16.3 FAQ on gold price ↓↑ / gold theft vs RBI liabilities and asset mismatch

Once RBI prints the notes, it is RBI's liability. IT doesn't matter if gold residing in RBI's vault is stolen or its (gold's) value is increased or decreased. RBI still has to exchange the notes with equivalent notes e.g. ₹500 = ₹100 x 5 Nos. How RBI will do it? That forensic analysis is not imp for our exam.



This is how 🧛 **vampire babushone** drink teacher's blood and yet **FAIL in prelims**. you must avoid this mentality. Move to next topic. what is the point of obsessing over a single topic like this?

11.17🖨️ Money Supply & FRBM Act



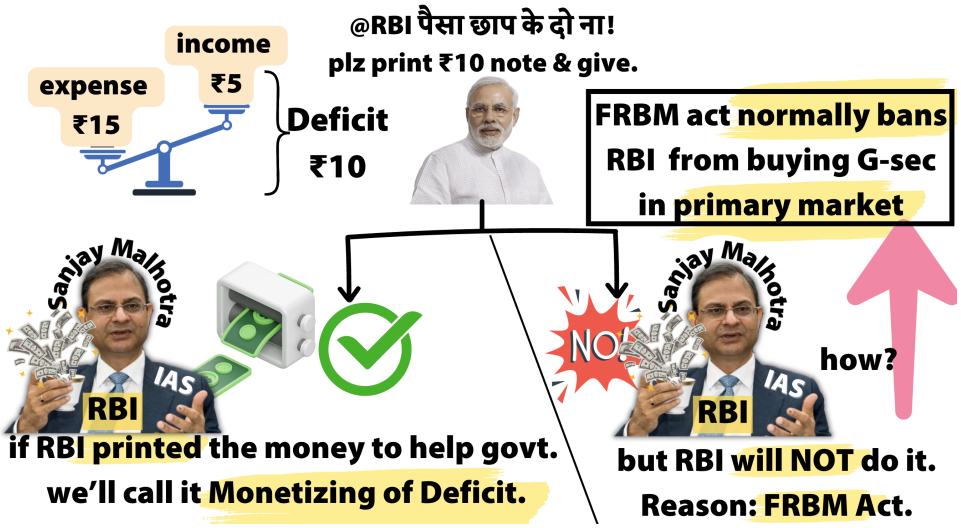
FISCAL RESPONSIBILITY AND BUDGET MANAGEMENT ACT 2003 (राजकोषीय उत्तरदायित्व और बजट प्रबंधन कानून) Notable provisions:

- Government has to control its fiscal deficit within X% of GDP. (Ref: Pillar#2D) (सरकार ने अपने घाटे को GDP के कुछ प्रतिशत से ज़्यादा नहीं बढ़ने देना होगा.)
- Government CANNOT borrow from RBI. (In other words, RBI cannot buy government's loan instruments for itself from primary market) **Except in TWO situations:**
    - i) Short term Ways and Means Advances (WMA) (Ref: Pillar#1C)
    - ii) During war, disaster, agri-crisis, big fall in GDP. (Ref: Pillar#2D)

- RBI can buy/sell government's loan instruments from secondary market (i.e. from other investors in OMO/ Open market operations.) (रिज़र्व बैंक आमतौर पे प्राथमिक बाज़ार से सरकार के लोन कर्ज़ उपकरण नहीं ख़रीद सकती है, किंतु द्वितीयक बाज़ार ले रिज़र्व बैंक उसकी ख़रीद बिक्री कर सकती है.)

🤦‍♀️**FAQ: didn't understand this.** Ans. Complete 1A2 pillar ke 5 lectures, Pillar1C AND pillar2D.

11.17.1 Monetisation of deficit (घाटे का मुद्रीकरण)



- It means RBI printing more currency notes to give as loans to Government. (जब सरकार को उधार देने के लिए रिजर्व बैंक नया पैसा छापती है)
- However, this process is difficult/ not easy due to FRBM Act conditions learned above. (हालाँकि FRBM क़ानून के चलते ये इतना आसान नहीं है अब.)

11.17.2 🚩👶 faaltu ke FAQ on Govt's borrowing & money printing

🚩👶FAQ: Can Govt borrow as much ₹₹ as they want via G-Sec? ✅ Ans. No! There are legal limits on govt borrowing under the Fiscal Responsibility And Budget Management act (FRBM). (More in 📄Pillar#2D) (सरकार अंधाधुंध तरीक़े से उधारी नहीं कर सकती FRBM क़ानून के अंतर्गत उस पर कुछ सीमाएं हैं.)

🚩👶FAQ: RBI's money supply is limited by above Asset-liability balance method. But what if govt issues large number of coins and one rupee notes before election to keep voters happy? Will it not create problems in the M0 & economy? ✅ Ans. Too much money supply without enough goods-supply → inflation. So, a sensible govt will not do such drama. Beyond that I've no interest in engaging in further PHD-intellectual debates. Not important for exam. Move to next topic.

🚩👶FAQ: How did RBI print money in British times? Ans.✅ Watch pill3B lecture videos

🚩👶FAQ: Where is the repo, where is the RBI's profit in above table? Ans.✅ Watch the videos

🚩👶FAQ: If RBI's gold/G-sec gets stolen/burned/lost then will M0 decrease? ✅ Watch the videos

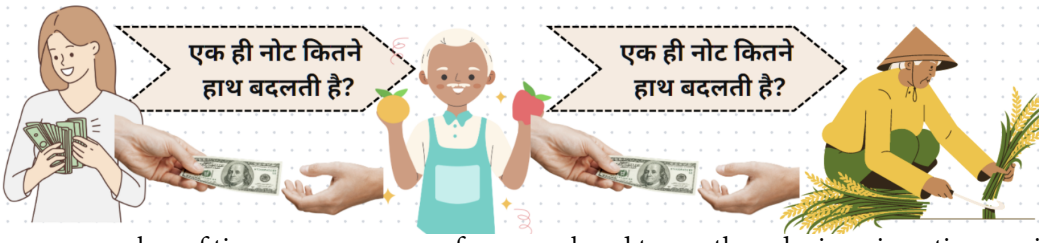
🚩👶FAQ: I've 500 jaat-ke other doubts. Ans.✅ then u keep wasting time on Google/ChatGPT.

11.17.3 💸⚡: Currency in Circulation (CIC)

- What is CIC's exact formula? Ans. Economic Survey didn't give, Google giving vague ans.

- CIC is an indicator to measure cash as a payment instrument i.e. instead of other payment instruments such as Cheque, NEFT/RTGS, Card etc. (भुगतान के लिए नक़दी का कितना प्रयोग हो रहा है)
- 2022: Russia-Ukraine war → temporary increase in ⬆️ precautionary cash holdings → % wise Currency in Circulation (CIC) increased (bcoz deposits were withdrawn).

11.17.4 💸 Misc. topic: Velocity of Money Circulation (मुद्रा संचालन का वेग)



It is the average number of times money passes from one hand to another, during given time period. e.g. you bought pen worth Rs.10 from shopkeeper, he uses same ₹10 note to buy tea from another shop → same currency note performed function of Rs.20. कितनी बार एक नोट एक हाथ से दूसरे हाथ में जाता है? This "Velocity of money circulation" is affected by following factors:

- **Income distribution:** Money in the hands of poor people has higher velocity than the rich people. Why/HOW? Ans. Watch Video.
- If more people borrow money → higher velocity. Hence developed countries => higher velocity, because people save less and spend more because of consumerist lifestyle and confidence in Government's social-security/pension system e.g. USA. (जब लोगों में बचत की भावना कम हो)
- Boom period in economy = more raw material purchase & hiring = higher velocity. (तेजी का माहौल)

Next Handout: 1A2-B: Monetary Policy (मौद्रिक नीति) - CRR, SLR, Repo, PSL etc.